



KINGDOM OF SAUDI ARABIA | JAZAN UNIVERSITY  
COLLEGE OF COMPUTER SCIENCE & INFORMATION  
SYSTEMS

Name of the Course: Elective-II (Java Programming )

**Project Title:**  
**Supermarket Management System**



| Student Name | Student ID |
|--------------|------------|
| Lolwa awada  | 202104346  |
| Basma Nasser | 202206835  |
| Radhya Hasan | 202210255  |
| SHAIMAA ALI  | 202200618  |

## **Content of table**

|                                               |   |
|-----------------------------------------------|---|
| 1. Introduction.....                          | 3 |
| 2. Problem Statement.....                     | 3 |
| 3. Technologies Used for Implementation ..... | 3 |
| 4. Input and Output Screens.....              | 4 |
| 5. Code .....                                 | 8 |

## 1. Introduction

The **Supermarket Management System** is a web-based platform designed to simplify inventory management tasks in a supermarket. It allows users to manage product details, such as adding, viewing, updating, and deleting products. The system also includes a login feature for secure access and a well-designed home page for navigation.

### Key Sections of the System:

- **Home:** Overview and navigation.
- **Add Products:** Add new products to the inventory.
- **View Products:** Display all products with details.
- **Edit Products:** Update product information.
- **About:** Information about the system.
- **Contact Us:** Contact details or form.
- **Logout:** Securely log out of the system.

## 2. Problem Statement

Supermarkets face challenges in managing inventory due to manual errors and inefficiencies. The **Supermarket Management System** aims to:

1. Provide a secure, accessible solution for inventory management.
2. Enable easy product addition, viewing, updating, and deletion.
3. Improve operational efficiency through a structured, interactive interface.

## 3. Technologies Used for Implementation

The **Supermarket Management System** follows a two-tier architecture with the following technologies:

## 1. Front-End:

- HTML for structure.
- CSS for styling.

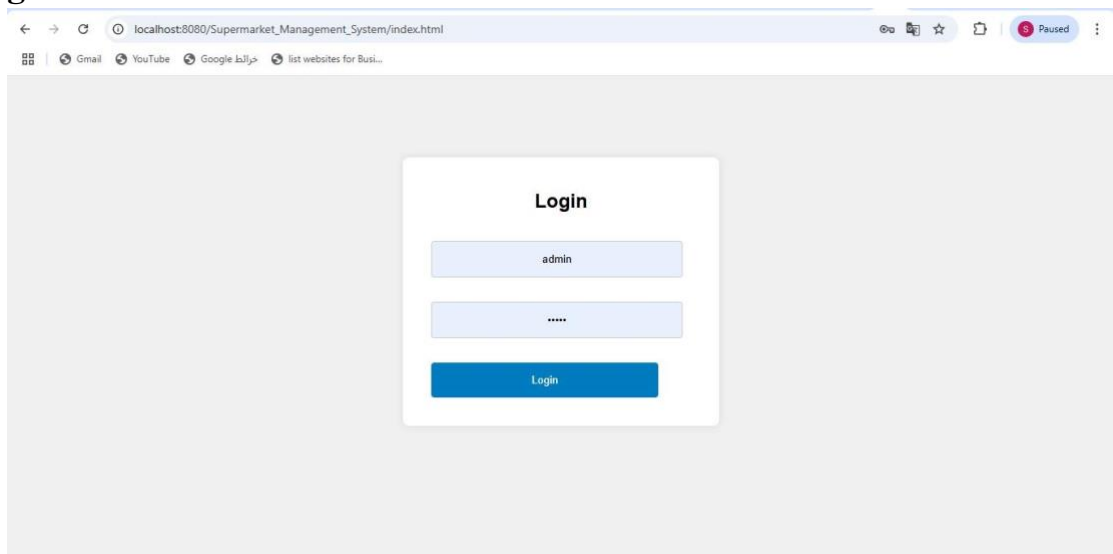
## 2. Back-End:

- Servlet for handling requests and responses.
- MySQL database for storing and retrieving data.

## 4. Input and Output Screens

### Input Screens:

#### Login:



localhost:8080/Supermarket\_Management\_System/index.html

Paused

Google Gmail YouTube Google Chrome list websites for Busi...

**Login**

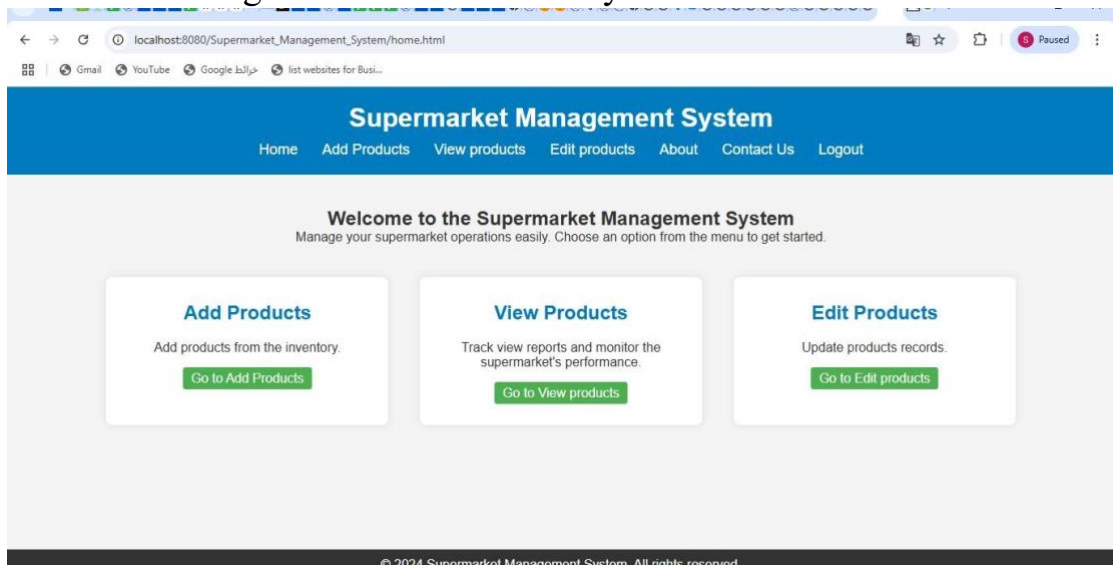
admin

\*\*\*\*\*

Login

### Home Page

- Central navigation with links to all system features:



## Add Products Page:

Input form to enter product details.

**Supermarket Management System**

Home Add Products View products Edit products About Contact Us Logout

**Manage Products**  
**Add New Product**

Product ID  
1010

Product Name  
Milk

Price:  
10.5

Quantity: Please enter a valid value. The two nearest valid values are 10 and 11.  
20

Add Product

## View Products

- Displays all products in a tabular format. □

Includes details such as:

**Supermarket Management System**

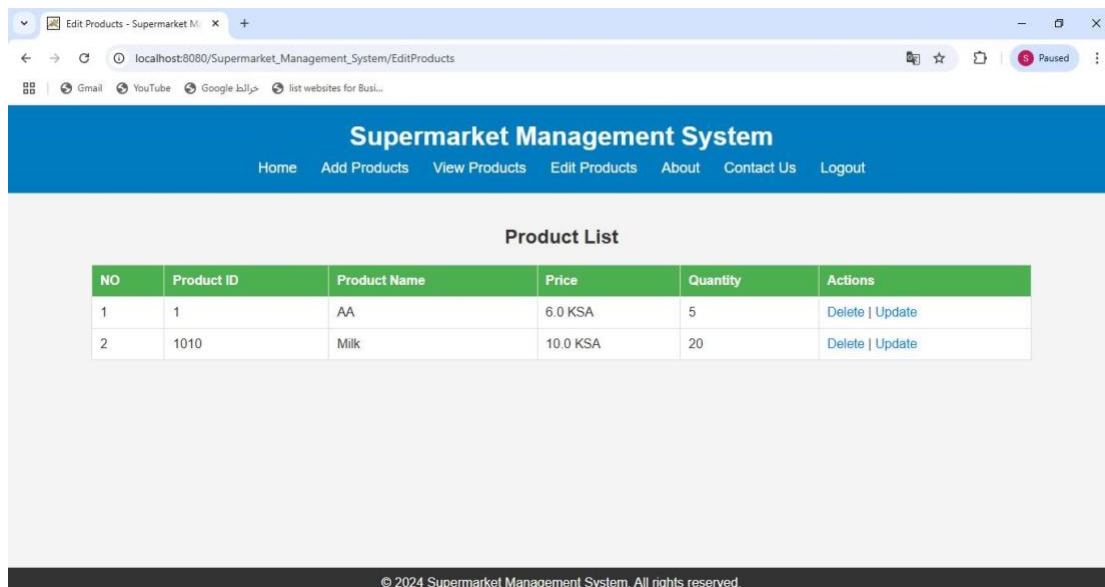
Home Add Products View Products Edit Products About Contact Us Logout

**Product List**

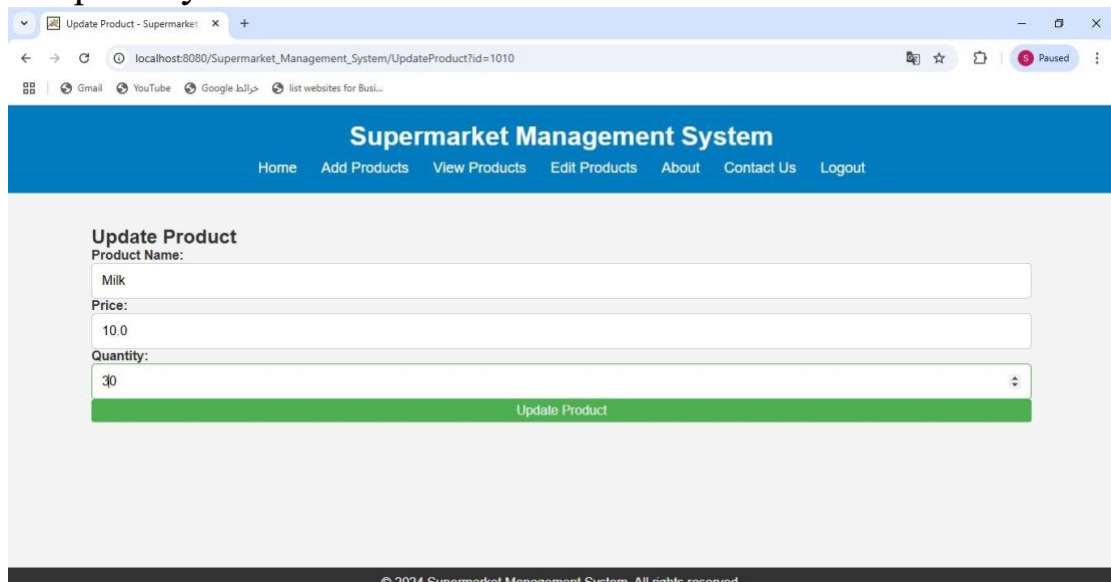
| NO | Product ID | Product Name | Price    | Quantity |
|----|------------|--------------|----------|----------|
| 1  | 1          | AA           | 6.0 KSA  | 5        |
| 2  | 1010       | Milk         | 10.0 KSA | 20       |

© 2024 Supermarket Management System. All rights reserved.

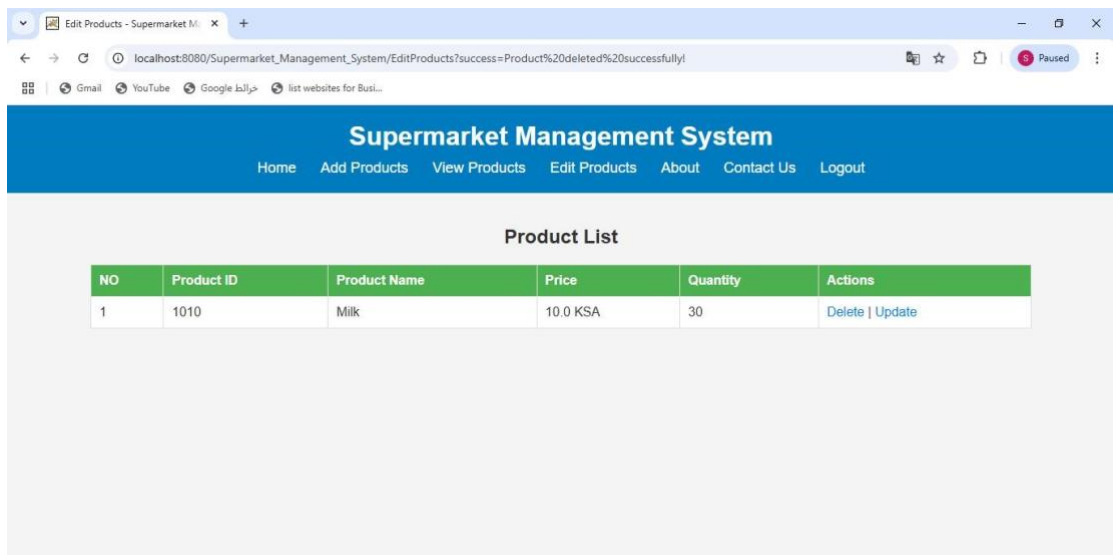
## Edit Product Page:



- **Update Product:**
  - Modify existing product information like name, price, and quantity.

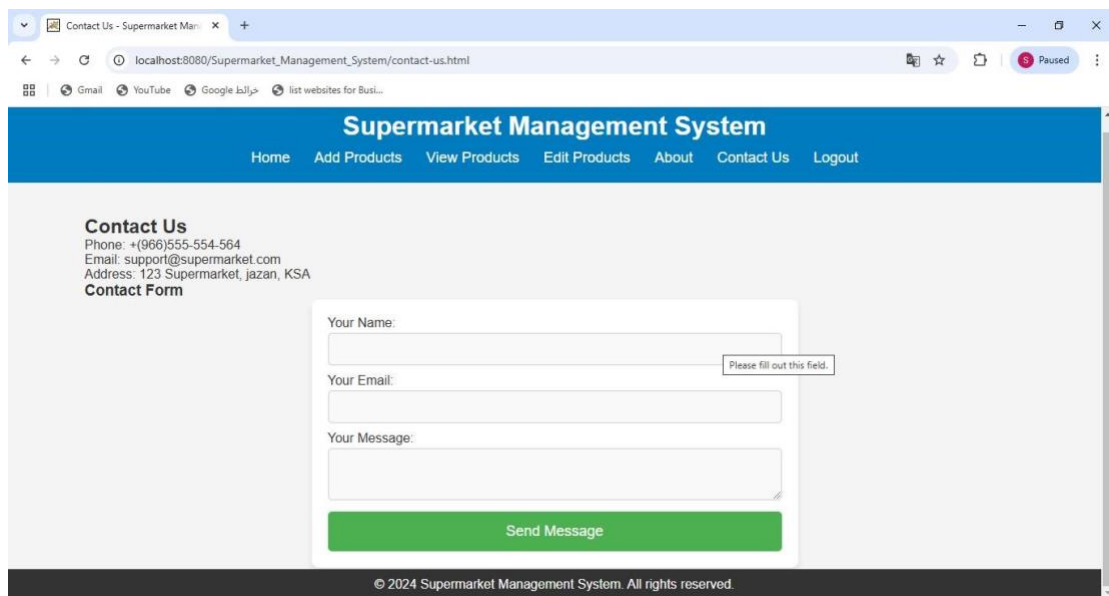


- **Delete Product:**
  - Permanently remove a product from the inventory.

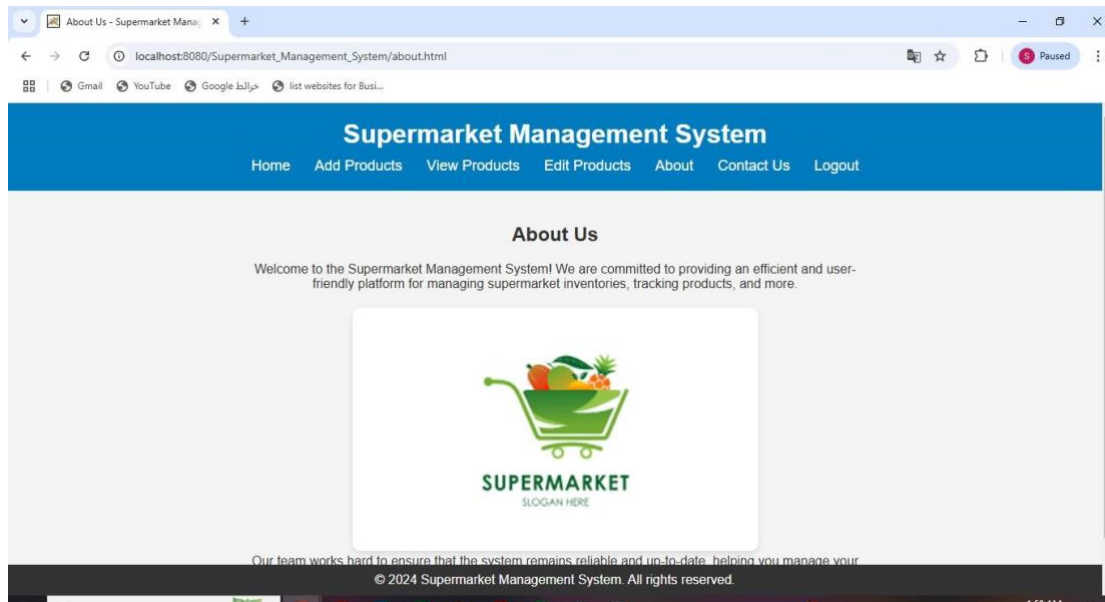


## Contact Us Form:

Fields for Name, Email, and Message.



## About



## 5. Code

### Index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initialscale=1.0">
  <title>Login - Student Information System</title>
</head>
<body>
  <div class="container">

    <h2>Login</h2>
    <form class="form" method="POST" action="LoginServlet">
      <input type="text" name="user_name" placeholder="Username"
required>
      <input type="password" name="user_password"
placeholder="Password" required>
      <button type="submit">Login</button>
    </form>
  </div>
</body>
</html>
```

### LoginServlet.java



```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import
java.sql.PreparedStatement;
import java.sql.ResultSet;
import javax.servlet.RequestDispatcher;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;

@WebServlet(urlPatterns = {"/LoginServlet"})
public class LoginServlet extends HttpServlet {
    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response) throws IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        String username = request.getParameter("user_name"); String
        password = request.getParameter("user_password"); try (Connection conn
        = DatabaseConnection.initializeDatabase()) {
```

```

        PreparedStatement stmt = conn.prepareStatement("SELECT *
FROM admin WHERE username = ? AND password = ?");
        stmt.setString(1, username);
        stmt.setString(2, password);
        ResultSet rs = stmt.executeQuery();
        if (rs.next()) {
            // Redirect to home if credentials are valid
            response.sendRedirect("home.html");
        } else {
            // Redirect back to login page with error message
            out.println("<center><h1><p style='color:red'>Error:
Invalid Login</p></h1></center>");
            RequestDispatcher dis =
request.getRequestDispatcher("index.html");
            dis.include(request, response);
        }
    } catch (Exception e) {
        e.printStackTrace();
        response.sendRedirect("index.html?error=Internal server
error");
    }
}

@Override
protected void doGet(HttpServletRequest request,
HttpServletRequest response) throws IOException {
    // Redirect to login page for GET requests
    response.sendRedirect("index.html"); }
}

```

## Home.html

```

<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initialscale=1.0">
    <title>Supermarket Management System - Home</title>
    <link rel="stylesheet" href="style.css">
</head>
<body>
    <header>
        <div class="header-container">
            <h1>Supermarket Management System</h1>
            <nav>
                <ul>
                    <li><a href="home.html">Home</a></li>
                    <li><a href="Add-products.html">Add
Products</a></li>

```

```

        <li><a href="viewProducts">View products</a></li>
        <li><a href="EditProducts">Edit
products</a></li> <li><a
href="about.html">About</a></li>
        <li><a href="contact-us.html">Contact Us</a></li>
        <li><a href="index.html">Logout</a></li>
    </ul>
</nav>
</div>
</header>

<main>
    <section class="welcome-section">
        <h2>Welcome to the Supermarket Management System</h2>
        <p>Manage your supermarket operations easily. Choose an
option from the menu to get started.</p>
    </section>
    <section class="features">
        <div class="feature">
            <h3>Add Products</h3>
            <p>Add products from the inventory.</p>
            <a href="Add-products.html" class="btn">Go to Add
Products</a>
        </div>
        <div class="feature">
            <h3>View Products</h3>
            <p>Track view reports and monitor the supermarket's
performance.</p>
            <a href="viewProducts" class="btn">Go to View
products</a>
        </div>
        <div class="feature">
            <h3>Edit Products</h3>
            <p>Update products records.</p>
            <a href="EditProducts" class="btn">Go to Edit
products</a>
        </div>
    </section>
</main>
<footer>
    <div class="footer-container">
        <p>&copy; 2024 Supermarket Management System. All rights
reserved.</p>
    </div>
</footer>
</body>
</html>

```

**DatabaseConnection.java**

```

import java.io.IOException; import
java.io.PrintWriter; import java.sql.Connection;
import java.sql.DriverManager; import
java.sql.SQLException; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/DatabaseConnection"})
public class DatabaseConnection extends HttpServlet
{
    public static Connection initializeDatabase() throws
        SQLException,
        ClassNotFoundException {
        String dbURL = "jdbc:derby://localhost:1527/Supermarket";
        String user = "root";
        String password = "password";
        Class.forName("org.apache.derby.jdbc.ClientDriver");
        return DriverManager.getConnection(dbURL, user, password); }
    @Override public String
    getServletInfo() { return "Short
    description";
    } // </editor-fold>
}

```

## AddProductServlet

```

import java.io.IOException; import
java.io.PrintWriter; import
java.sql.Connection; import
java.sql.PreparedStatement; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;

@WebServlet("/AddProductServlet") public class
AddProductServlet extends HttpServlet { protected
void doPost(HttpServletRequest request,
HttpServletResponse response) throws ServletException, IOException {
    PrintWriter out= response.getWriter();
}
}

```

```

        String productID = request.getParameter("product_id");
        String productName = request.getParameter("product_name");
        String productPrice = request.getParameter("product_price");
        String productQuantity =
request.getParameter("product_quantity");

        try (Connection conn =
DatabaseConnection.initializeDatabase())
{
            String query = "INSERT INTO products (id, name, price,
quantity) VALUES (?, ?, ?, ?)";
            PreparedStatement stmt = conn.prepareStatement(query);
            stmt.setString(1, productID); stmt.setString(2,
productName); stmt.setString(3, productPrice);
            stmt.setString(4, productQuantity);
            out.println("<center><h1><p style= color:red>Data has
been
stored</p></h1> </center>"); int rowsAffected =
            stmt.executeUpdate(); if (rowsAffected
            > 0) {
                response.sendRedirect("Add-
products.html?success=Product added successfully!");
            } else {
                response.sendRedirect("Add-products?error=Failed to add
product.");
            }
        } catch (Exception e) {
            e.printStackTrace(); response.sendRedirect("Add-
products?error=An error occurred
while adding the product.");
        }
    }
}

```

## DeleteProductServlet

```

import java.io.IOException; import
java.sql.Connection; import
java.sql.PreparedStatement; import
javax.servlet.ServletException; import
javax.servlet.annotation.WebServlet; import
javax.servlet.http.HttpServlet; import
javax.servlet.http.HttpServletRequest; import
javax.servlet.http.HttpServletResponse;

@WebServlet("/DeleteProductServlet") public class
DeleteProductServlet extends HttpServlet { protected
void doGet(HttpServletRequest request,
HttpServletResponse response) throws ServletException, IOException {

```

```

String productId = request.getParameter("id");

try (Connection conn = DatabaseConnection.initializeDatabase())
{
    String query = "DELETE FROM products WHERE id = ?";
    PreparedStatement stmt = conn.prepareStatement(query);
    stmt.setInt(1, Integer.parseInt(productId));
    int rowsAffected = stmt.executeUpdate();
    if (rowsAffected > 0) {
        response.sendRedirect("EditProducts?success=Product
deleted successfully!");
    } else {
        response.sendRedirect("EditProducts?error=Failed to
delete product.");
    }
} catch (Exception e) {
    e.printStackTrace();
    response.sendRedirect("EditProducts?error=An error
occurred
while deleting the product.");
}
}
}

```